

$$A. TOTALSCORE_i = 918.628 + 9.989 \times SMALL_i$$

Table 1: Effect of Class Size on Student Aptitude

	Dependent variable:
	Total Score
Small Class	9.989* (4.611)
Constant	918.628*** (2.536)
Observations	1,200
R ²	0.004
Adjusted R ²	0.003
Residual Std. Error	73.364 (df = 1198)
F Statistic	4.694* (df = 1; 1198)
Note: *p<0.05, **p<0.01, ***p<0.001	

The regression results indicate a positive relationship between smaller class sizes and student achievement. The coefficient for the small class variable is 9.989 with a p-value of 0.0305, which is statistically significant at the 5% level. Because the Project STAR experiment utilized random assignment to place students into class types, this experimental design eliminates selection bias. Therefore, we can confidently conclude that reducing class size to 13–17 students has a positive, causal effect on early childhood learning and test scores.

B. $READSCORE_i = 434.123 + 5.466 \times SMALL_i$

Table 2: Effect of Class Size on Student Aptitude

	Dependent variable:
	Read Score
Small Class	5.466** (1.897)
Constant	434.123*** (1.044)
Observations	1,200
R ²	0.007
Adjusted R ²	0.006
Residual Std. Error	30.190 (df = 1198)
F Statistic	8.301** (df = 1; 1198)
Note:	*p<0.05, **p<0.01, ***p<0.001

$MATHSCORE_i = 484.505 + 4.522 \times SMALL_i$

Table 3: Effect of Class Size on Student Aptitude

	Dependent variable:
	Math Score
Small Class	4.522 (3.049)

Constant	484.505*** (1.677)
Observations	1,200
R ²	0.002
Adjusted R ²	0.001
Residual Std. Error	48.522 (df = 1198)
F Statistic	2.199 (df = 1; 1198)
Note:	*p<0.05, **p<0.01, ***p<0.001

Yes, there is a clear difference in how class size affects reading versus math scores.

1. *Reading Score (READSCORE):*

The regression shows that small classes have a statistically significant positive effect on reading scores. The estimated coefficient is 5.466 with a p-value of 0.00403 ($p < 0.01$). This indicates that students in small classes score about 5.5 points higher in reading than those in regular classes, and this result is highly significant.

2. *Math Score (MATHSCORE):*

In contrast, the effect of small classes on math scores is not statistically significant. Although the estimated coefficient is positive (4.522), the p-value is 0.138, which is greater than the standard 0.05 significance level. We fail to reject the null hypothesis, meaning we cannot confidently say that small classes improve math scores.

Conclusion

The primary difference observed is that class size significantly impacts early childhood reading skills, but not math skills. Furthermore, because TOTALSCORE is a combination of these two subjects, the overall significance found in part (a) ($p = 0.0305$) is almost entirely driven by the strong improvement in reading. The lack of significance in the math scores essentially

dilutes the stronger effect seen in the reading scores when they are combined.

$$C. \text{TOTALSCORE}_i = 922.145 - 1.396 \times \text{AIDE}_i$$

Table 4: Effect of AIDE on Student Aptitude

<i>Dependent variable:</i>	
	Total Score
AIDE	-1.396 (4.437)
Constant	922.145*** (2.640)
Observations	1,200
R ²	0.0001
Adjusted R ²	-0.001
Residual Std. Error	73.504 (df = 1198)
F Statistic	0.099 (df = 1; 1198)
<i>Note:</i> *p<0.05, **p<0.01, ***p<0.001	

The regression results demonstrate that adding a full-time teacher aide to a regular-sized classroom has **no statistically significant effect** on student learning. Although the estimated coefficient is slightly negative (-1.396), its p-value is 0.753, which is substantially larger than the standard 0.05 significance level. Consequently, we fail to reject the null hypothesis. We can confidently conclude that the presence of a teacher aide does not improve students' early childhood academic achievement.

$$D. \text{READSCORE}_i = 435.908 - 0.372 \times \text{AIDE}_i$$

Table 5: Effect of AIDE on Student Aptitude

<i>Dependent variable:</i>	
	Read Score
AIDE	-0.372 (1.829)
Constant	435.908*** (1.088)
Observations	1,200
R ²	0.00003
Adjusted R ²	-0.001
Residual Std. Error	30.294 (df = 1198)
F Statistic	0.041 (df = 1; 1198)
Note:	*p<0.05, **p<0.01, ***p<0.001

$$MATHSCORE_i = 486.236 - 1.024 \times AIDE_i$$

Table 6: Effect of AIDE on Student Aptitude

<i>Dependent variable:</i>	
	Math Score
AIDE	-1.024 (2.931)
Constant	486.236***

(1.744)

Observations	1,200
R ²	0.0001
Adjusted R ²	-0.001
Residual Std. Error	48.564 (df = 1198)
F Statistic	0.122 (df = 1; 1198)
Note:	*p<0.05, **p<0.01, ***p<0.001

There are no meaningful differences observed between the two subjects in this case. The presence of a teacher aide has no statistically significant effect on either reading or math scores.

1. *Reading Score (READSCORE):*

The estimated coefficient for the aide variable is -0.3719 with a p-value of 0.839. Since the p-value is much greater than 0.05, we fail to reject the null hypothesis. Having a teacher aide does not significantly affect reading scores.

2. *Math Score (MATHSCORE):*

Similarly, the estimated coefficient for the aide variable on math scores is -1.024 with a p-value of 0.727. This is also not statistically significant.

Conclusion:

Unlike in part (b), where class size reduction had a significant impact on reading but not math, adding a teacher aide fails to improve test scores in *either* subject. Both coefficients are slightly negative but statistically indistinguishable from zero. This consistent lack of significance across both reading and math perfectly explains why the combined TOTALSCORE in part (c) also showed no significant improvement. The results suggest that hiring a teacher aide for a regular-sized classroom does not provide the same educational benefits as reducing the actual class size.

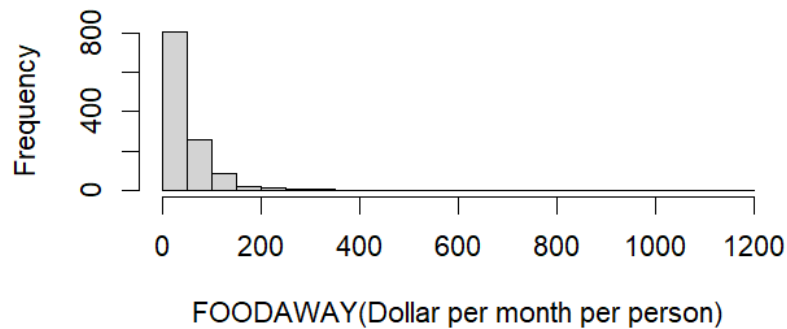
2.25

A. Construct a histogram of FOODAWAY and its summary statistics. What are

the mean and median values? What are the 25th and 75th percentiles?

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	12.04	32.55	49.27	67.50	1179.00

Histogram of foodaway



- B. What are the mean and median values of FOODAWAY for households including a member with an advanced degree? With a college degree member? With no advanced or college degree member?

N (Advanced): 257

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	21.67	48.15	73.15	90.00	1179.00

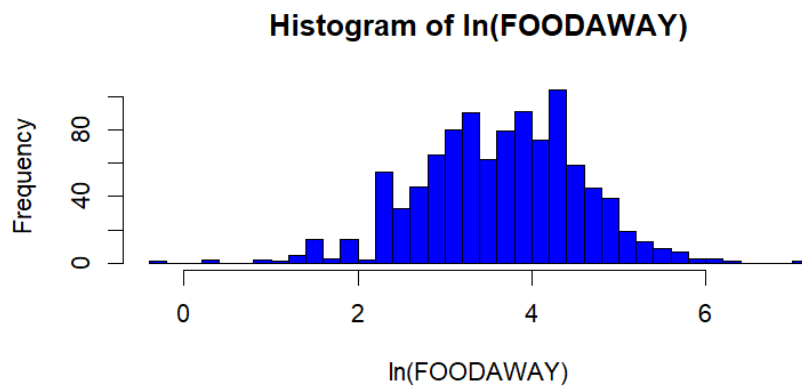
N (College): 369

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	14.44	36.11	48.60	68.67	416.11

N (Non-college and Non-advanced): 574

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	9.63	26.02	39.01	52.65	437.78

- C. Construct a histogram of $\ln(\text{FOODAWAY})$ and its summary statistics. Explain why FOODAWAY and $\ln(\text{FOODAWAY})$ have different numbers of observations.



N (Observed): 1200

N (Observed without nan): 1022

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.3011	3.0759	3.6865	3.6508	4.2797	7.0724

Because $\ln(0)$ is undefined, FOODAWAY and $\ln(\text{FOODAWAY})$ have different numbers of observations.

- D. Estimate the linear regression $\ln(\text{FOODAWAY}) = \beta_1 + \beta_2 \text{INCOME} + e$.

Interpret the estimated slope.

$$\ln(\text{FOODAWAY})_i = 3.129 + 0.007 \times \text{INCOME}_i$$

Table 7

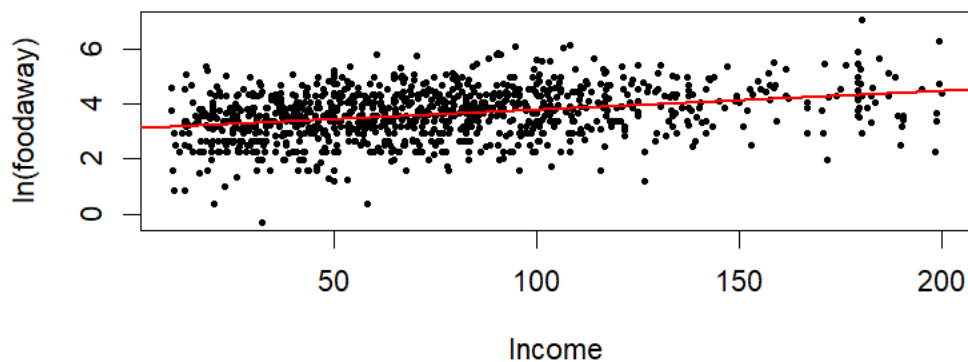
<i>Dependent variable:</i>	
ln(foodaway)	
INCOME	0.007*** (0.001)
Constant	3.129*** (0.057)
Observations	1,022
R ²	0.098
Adjusted R ²	0.097
Residual Std. Error	0.876 (df = 1020)
F Statistic	111.150*** (df = 1; 1020)

*p<0.05, **p<0.01, ***p<0.001

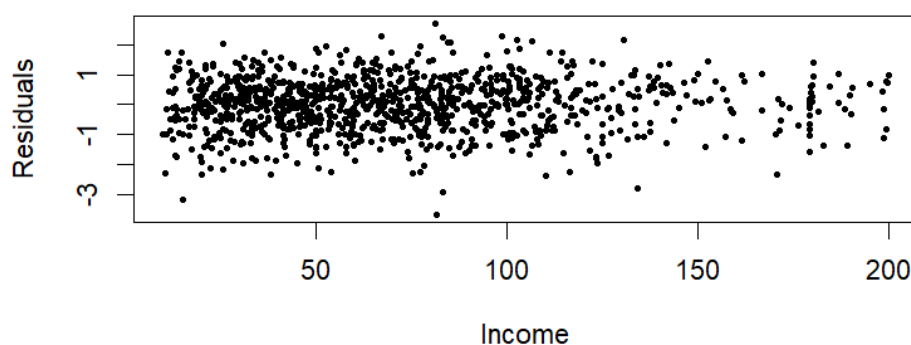
The estimated coefficient for income is 0.0069. Because the dependent

variable is log-transformed, the interpretation of the slope shifts to a percentage change. Specifically, a one-unit increase in household income is associated with an approximate 0.69% increase in food away from home spending

- E. Plot $\ln(\text{FOODAWAY})$ against INCOME , and include the fitted line from part (d).



- F. Calculate the least squares residuals from the estimation in part (d). Plot them vs. INCOME . Do you find any unusual patterns, or do they seem completely random?



Based on the visual inspection of the residual plot (Residuals versus Income), the residuals appear to be randomly and evenly scattered around the horizontal zero line across all levels of household income. We do not observe any systematic or predictable patterns, such as a funnel or fan shape, which would suggest that the variance of the error term changes as income

increases. Therefore, we can conclude that the model satisfies the assumption of homoskedasticity (constant variance of residuals). Because this assumption holds, the Ordinary Least Squares (OLS) standard errors are valid, meaning the t-statistics and p-values reported in our regression model are statistically reliable for hypothesis testing.